

Lead Management System (LMS) - Database Architecture & UI/UX Design System

Purpose:

This document defines the complete database planning and user interface design standards required to build an enterprise-level Lead Management System.

This document should be followed by Database Engineers and UI/UX Designers to keep the complete product consistent.

PART 1: DATABASE ARCHITECTURE

Database Technology:

MySQL

Architecture Goal:

Create a scalable, normalized and production-ready database structure.

1. Database Tables Overview

Core Tables:

users

roles

permissions

role_permissions

leads

lead_sources

lead_status

followups

lead_documents

lead_notes

lead_activity_logs

notifications

2. Users Table

Purpose:
Stores system users.

Columns:

id

name

email

password

phone

role_id

status

created_at

updated_at

Relationship:

User belongs to Role

User has many Leads

3. Roles & Permissions Tables

roles:

id

role_name

permissions:

id

permission_name

role_permissions:

role_id

permission_id

Purpose:

Controls Role Based Access Control system.

4. Leads Table

Main CRM entity.

Columns:

id

name

company

email

phone

address

industry

source_id

campaign

product

budget

priority

status_id

assigned_user_id

followup_date

created_by

created_at

updated_at

Relationships:

Lead belongs to User

Lead has many Followups

Lead has many Notes

Lead has many Documents

5. Followups Table

Stores customer interaction planning.

Columns:

id

lead_id

user_id

type

date

time

status

remarks

Types:

Call

Meeting

Email

WhatsApp

Visit

6. Documents & Notes

lead_documents:

Stores uploaded files.

lead_notes:

Stores communication notes and internal remarks.

7. Activity Logs

Purpose:

Maintain complete history.

Tracks:

Lead creation

Status changes

Assignments

Updates

Follow-ups

8. Database Optimization Rules

Requirements:

Use indexes

Use foreign keys

Maintain relationships

Avoid duplicate data

Support soft deletes

Store timestamps

PART 2: UI/UX DESIGN SYSTEM

Design Goal:

Create a modern enterprise CRM experience similar to professional SaaS products.

9. UI Inspiration

Inspired By:

Salesforce CRM

HubSpot CRM

Monday CRM

Modern SaaS Dashboards

Design Style:

Clean

Professional

Minimal

Data Focused

10. Application Layout

Main Layout:

Left Sidebar Navigation

Top Header

Content Area

Dashboard Widgets

Tables

Charts

Sidebar Items:

Dashboard

Leads

Follow-ups

Reports

Users

Settings

11. Dashboard UI Requirements

Dashboard Components:

Statistics Cards

Performance Charts

Lead Funnel

Recent Activities

Follow-up Reminders

Charts:

Lead Source

Monthly Growth

Conversion Ratio

Sales Performance

12. Component Design System

Reusable Components:

Buttons

Inputs

Dropdowns

Cards

Tables

Modals

Badges

Charts

Forms

Tabs

Pagination

Notifications

13. Lead Management UX

Lead List:

Search

Advanced Filters

Sorting

Pagination

Bulk Actions

Lead Details:

Overview

Timeline

Notes

Documents

Communication History

14. Design States

Every screen must include:

Loading State

Empty State

Error State

Success Feedback

Validation Messages

15. Responsive Design

Support:

Desktop

Laptop

Tablet

Mobile

Rules:

Adaptive Sidebar

Responsive Tables

Mobile Friendly Forms

16. Final Product Quality Rules

The final LMS should feel like:

Enterprise CRM

Production SaaS Application

Real Company Internal Tool

Focus:

Performance

Consistency

Accessibility

User Experience